



Experimental Process and Model Evaluation

Comparing Classifiers

Say we have two classifiers, $C1$ and $C2$, and want to choose the best one to use for future predictions

Can we use training accuracy to choose between them?

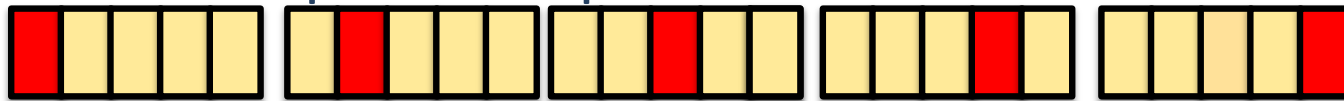
- No!
- What about accuracy on test data?
- Yes, but...
 - We basically want to look at more than a single number; gather some statistical evidence.

N-fold cross validation

- Instead of a single training-validation split:

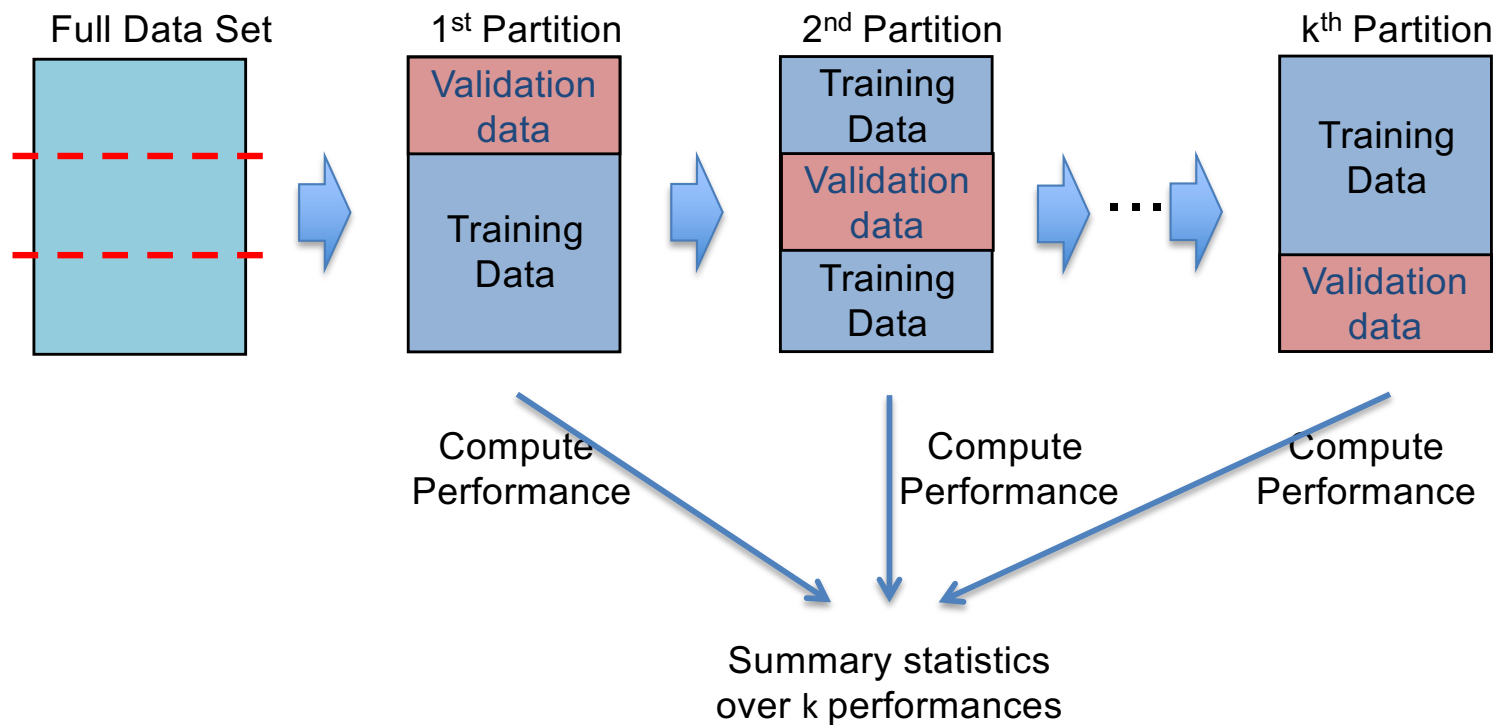


- Split data into N equal-sized parts



- Train and validate N different classifiers
- Report average accuracy and standard deviation of the accuracy

Example K-Fold CV



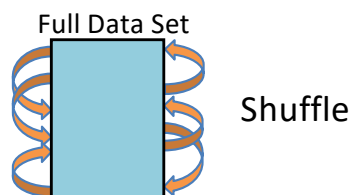
More on Cross-Validation

- Cross-validation generates an approximate estimate of how well the classifier will do on “unseen” data
 - As $k \rightarrow n$, the model becomes more accurate (more training data)
 - ...but, CV becomes more computationally expensive
 - Choosing $k < n$ is a compromise. $k=5$, or 10 is often used.
 - $k=n$ is called “leave-one-out”;
- Averaging over different partitions is more robust than just a single train/validate partition of the data
- It is an even better idea to do CV repeatedly!

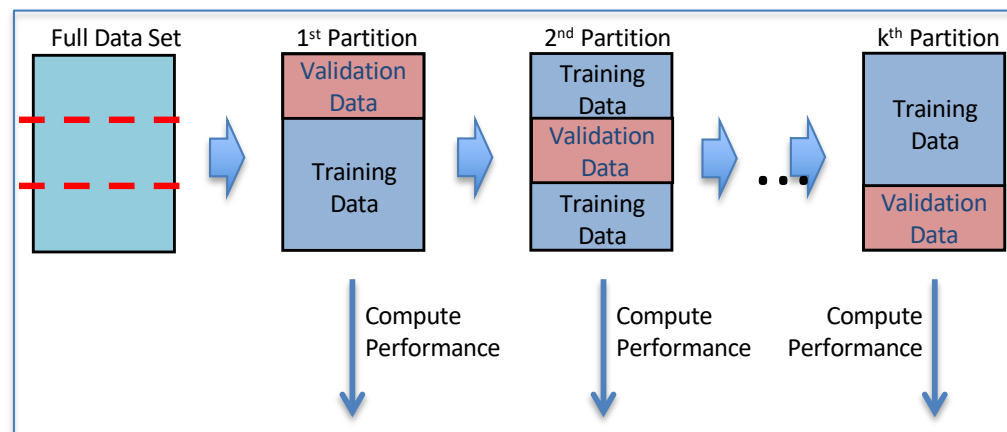
Multiple Trials of k-Fold CV

1.) Loop for t trials:

a.) Randomize
Data Set



b.) Perform
k-fold CV

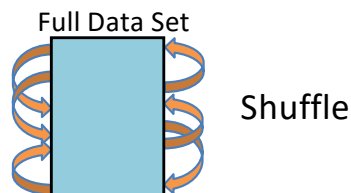


2.) Compute statistics over
 $t \times k$ test performances

Comparing Multiple Classifiers

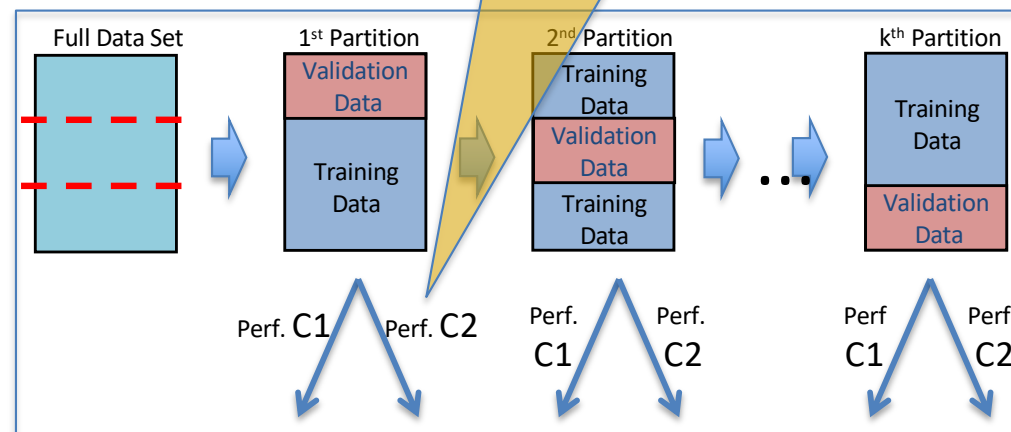
1.) Loop for t trials:

a.) Randomize Data Set



Test each candidate learner on same training/validation splits

b.) Perform k-fold CV



2.) Compute statistics over $t \times k$ test performances

Allows us to do paired summary statistics (e.g., paired t-test)

Evaluation: significance tests

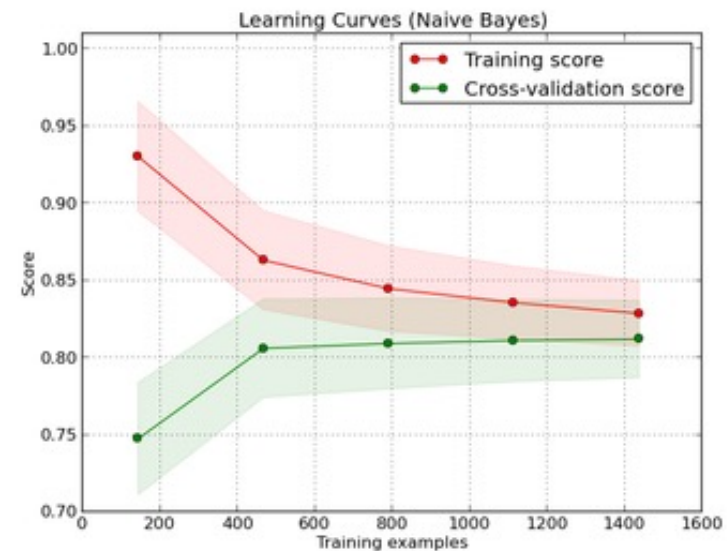
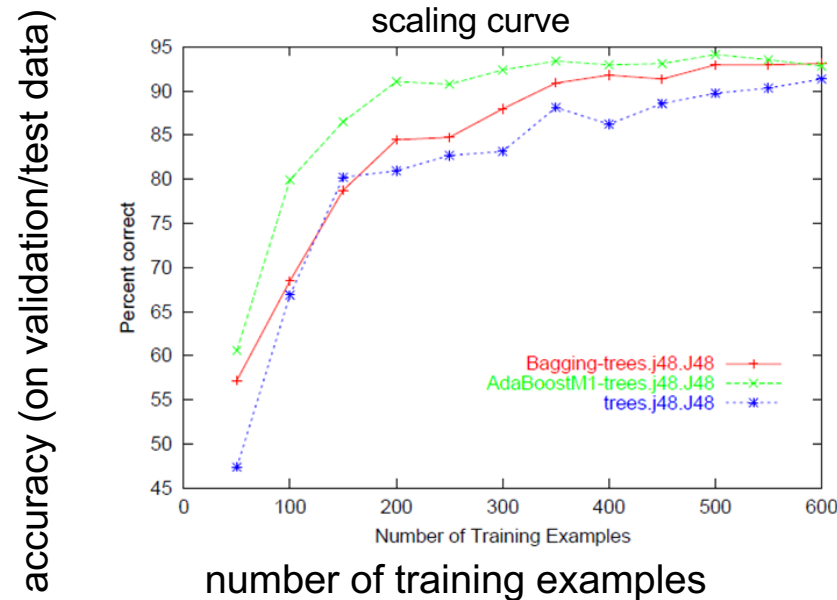
- You have two different classifiers, A and B
- You train and validate them on the same data set using N-fold cross-validation
- For the n -th fold:
 $\text{accuracy}(A, n), \text{accuracy}(B, n)$
 $p_n = \text{accuracy}(A, n) - \text{accuracy}(B, n)$
- Is the difference between A and B's accuracies significant?



— E.g., paired t-test

Learning Curve

- Shows performance versus the # training examples
 - Compute over a single training/validation split
 - Then, average across multiple trials of CV

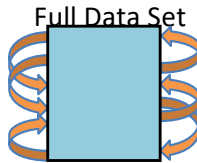


Building Learning Curves

`train_sizes = [100, 300, 500, 1000, 1500]`

1.) Loop for t trials, use `training_size[t]`:

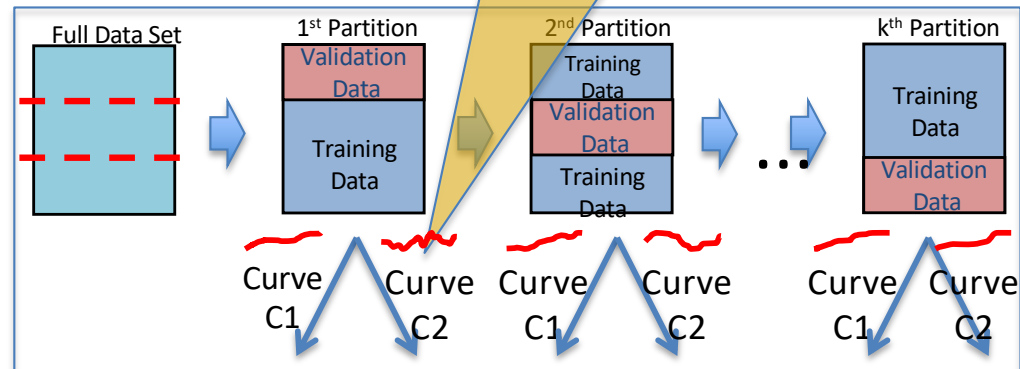
a.) Randomize
Data Set



Shuffle

Compute learning curve over each
training/testing split

b.) Perform
k-fold CV



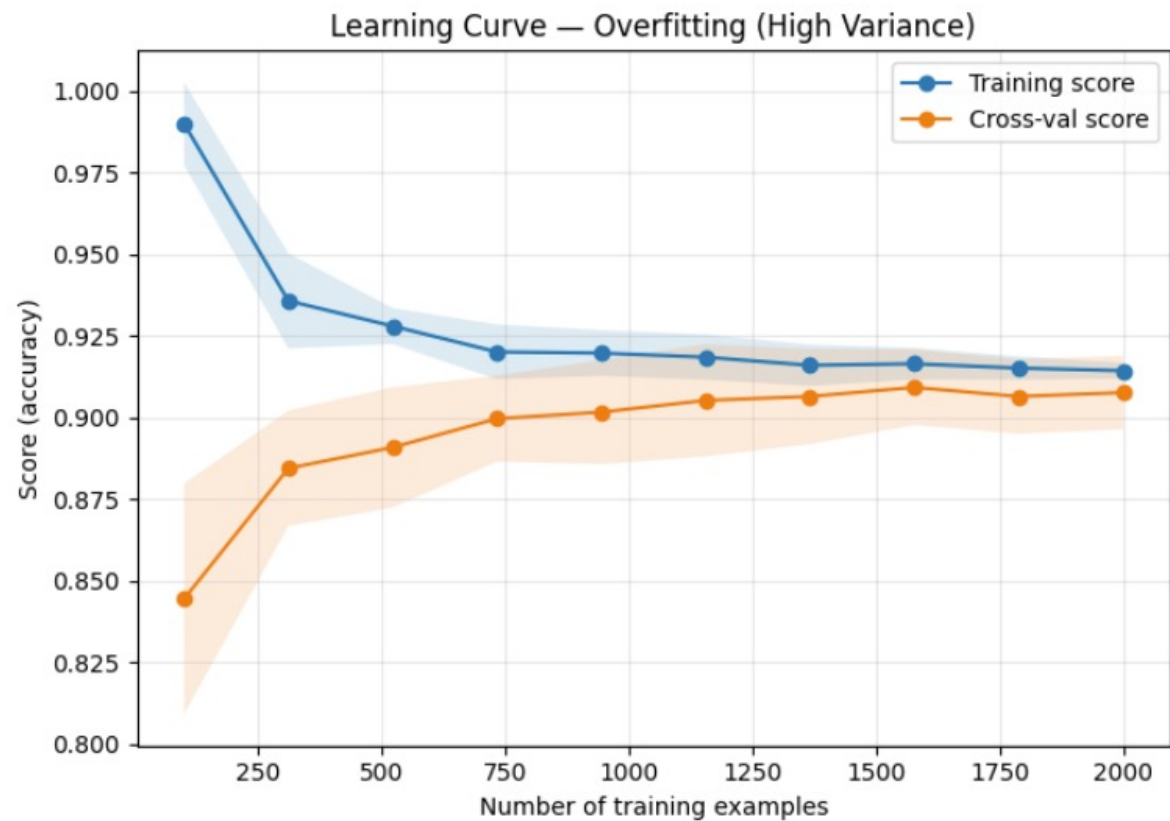
2.) Compute statistics over
 $t \times k$ learning curves

Learning Curve Characteristics

- **Training score**
 - Usually **starts high**
 - **Decreases slightly** as more data is added
 - Then stabilizes. Reason: harder to fit more diverse data
- **Validation score**
 - Starts **low**
 - **Increases** with more data
 - Then plateaus. Reason: better generalization
- **The gap between training and validation shrinks**

Learning Curve-- Overfitting

- **Overfitting:**
 - Training: high
 - Validation: low
 - Big gap
 - More data helps → gap shrinks



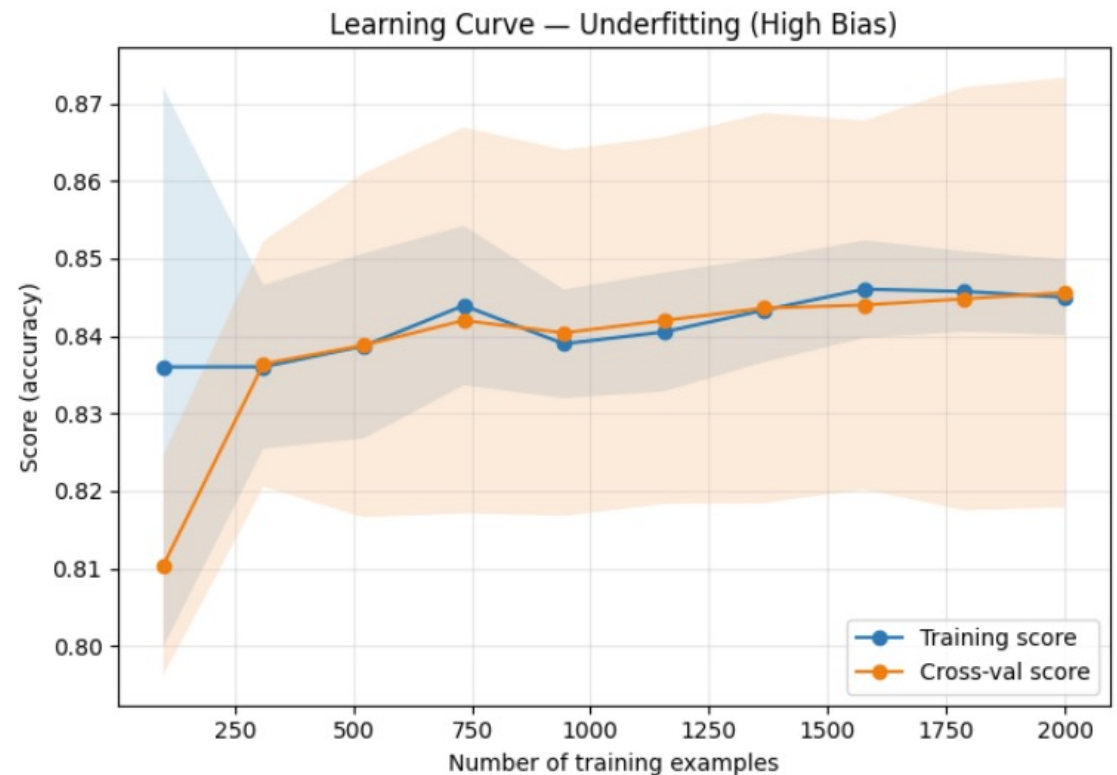
Learning Curve-- Underfitting

- **Underfitting**

- Both (relative) low

- No gap

- More data does NOT help



Learning Curve– Better fit

- **Small** and stable gap (key signal)
Model generalizes well (low variance)
- **Both curves are reasonably high**
model is learning meaningful patterns
not too simple
- **Flat** → limit reached

Good learning curve = high performance + small gap + stable plateau

